
EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 3A

1. Signal Fundamental Frequency

For each of the discrete-time signals below, determine the fundamental period (and corresponding fundamental frequency), or explain why the signal is not periodic.

(a) $x[n] = e^{i\frac{\pi}{5}n}$ for all integers n .

(b) $x[n] = \cos\left(\frac{3\pi}{5}n\right)$ for all integers n .

(c) $x[n] = \sin(n)$ for all integers n .

2. Orthogonal Building Blocks

For an integer k , define the vector $\phi_k \in \mathbb{C}^N$ by

$$\phi_k[n] = e^{-ik\frac{2\pi}{N}n}, \quad n = 0, 1, \dots, N-1.$$

Let's take $N = 4$, so each ϕ_k has 4 entries. Recall the inner product of two such vectors: $a, b \in \mathbb{C}^4$ as $\langle a, b \rangle = \sum_{n=0}^3 a[n] b^*[n]$.

- (a) Write out the 4 entries of $\phi_0, \phi_1, \phi_2, \phi_3$, then compute $\langle \phi_0, \phi_1 \rangle$ and $\langle \phi_1, \phi_3 \rangle$. What do you notice?

Hint: Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, will help you evaluate each entry.

- (b) Explain why $\phi_0, \phi_1, \phi_2, \phi_3$ can be used to construct *any* vector in $\mathbb{C}^4$ as a linear combination of them.